C^2-STVSR: Correlation-guided Continuous Spatial-Temporal Video Super-Resolution

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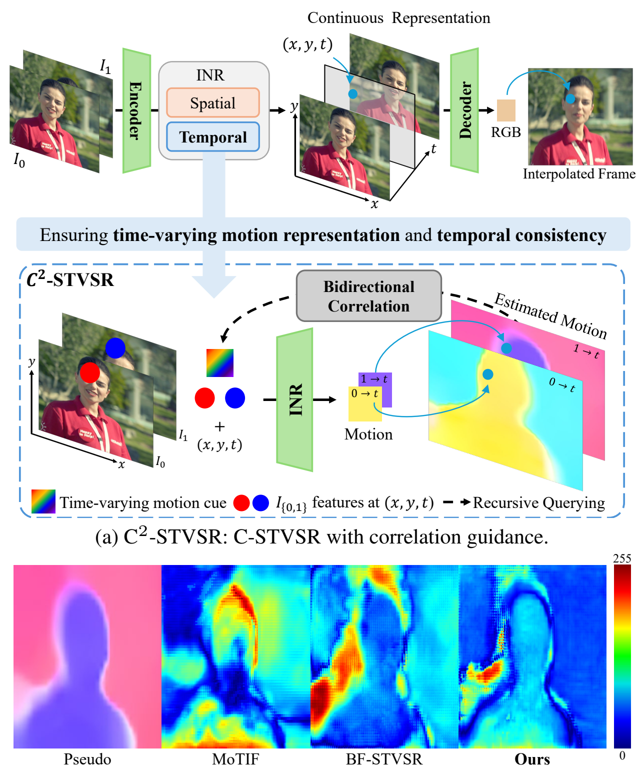
Abstract

Continuous Spatial-Temporal Video Super-Resolution (C-STVSR) aims to reconstruct high-resolution frames at arbitrary spatial and temporal scales with a single model. Recent INR-based methods have enabled this by modeling videos as continuous functions over spatial-temporal coordinates. Existing methods estimate motion, warp high-resolution reference features according to the estimated motion, and decode them to synthesize target frames, making reconstruction quality highly dependent on motion estimation accuracy. Despite this strong dependence, they typically estimate motion for each target time step independently by varying only a scalar temporal coordinate t, which results in motion multiplicity and temporal discrepancy. To address these limitations, we propose C^2-STVSR, which exploits bidirectional 4D correlation volumes to provide time-varying motion cues and enable recursive motion propagation, leading to temporally coherent motion representations. Extensive experiments on multiple benchmarks demonstrate that our method achieves state-of-the-art performance, producing sharper details and more temporally consistent reconstructions. Our code is available here.

1. Introduction

High-resolution and high-frame-rate videos are crucial for high-quality visual experiences in modern display environments. To satisfy this demand, deep learning-based Video Super-Resolution (VSR) and Video Frame Interpolation (VFI) have been widely studied to enhance spatial and temporal resolution, respectively. While both tasks have achieved remarkable progress, many real-world applications require simultaneous enhancement of spatial and temporal resolution. This has motivated research on Spatial-Temporal Video Super-Resolution (STVSR), which aims to jointly restore spatial detail and temporal dynamics. A common strategy is to cascade VFI and VSR models independently, but such two-stage pipelines fail to fully exploit the intrinsic coupling between spatial detail and temporal dynamics. Moreover, most existing methods are designed for fixed spatial and temporal scaling factors, limiting their flexibility in diverse real-world scenarios.



(a) C^2-STVSR: C-STVSR with correlation guidance.

(b) Difference map based on pseudo motion.

Figure 1. Illustration of C^2-STVSR and comparisons of estimated motion. (a) C^2-STVSR leverages a bidirectional 4D correlation volume to capture dynamic motion cues and propagate motion across time, enabling temporally coherent motion modeling for video reconstruction. (b) Difference map between the predicted motion and the pseudo motion estimated using RAFT [18].

Recently, Implicit Neural Representation (INR)-based methods have emerged as a promising approach for continuously modeling spatial and temporal dimensions, enabling Continuous Spatial-Temporal Video Super-Resolution (C-

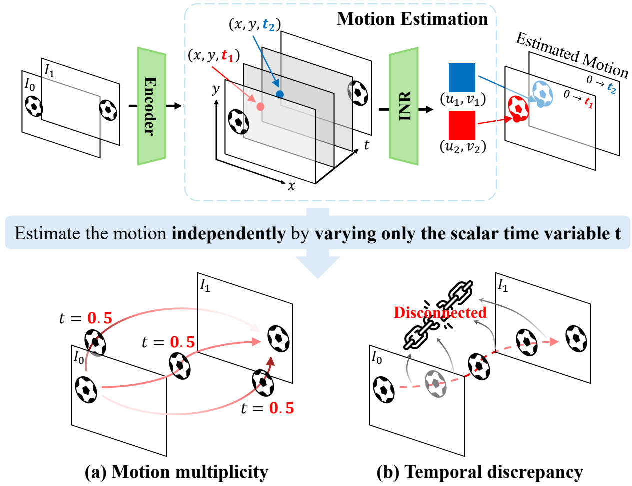


Figure 2. Limitations of existing C-STVSR motion modeling. (a) **Motion multiplicity:** Multiple plausible motion trajectories may exist between two frames, but varying only a scalar temporal variable t while keeping the reference feature maps fixed constrains the model to a one-dimensional temporal representation, collapsing diverse trajectories into an averaged motion pattern. (b) **Temporal discrepancy:** Estimating motion independently at each time step without incorporating prior motion information fails to enforce temporal continuity, leading to temporally inconsistent motion estimates.

STVSR) within a unified framework. VideoINR [5] first demonstrated the feasibility of INR-based C-STVSR by predicting motion from spatial-temporal coordinates, warping reference features to arbitrary coordinates, and decoding RGB values accordingly. MoTIF [4] improves motion trajectory modeling via forward warping with Softmax splatting [16] and a pre-trained optical flow network, RAFT [18]. BF-STVSR [11] addresses the spectral bias of INR-based C-STVSR by introducing axis-wise positional encoding with B-spline and Fourier features. Despite these advances, existing methods share a common pipeline: they predict motion at arbitrary spatial-temporal coordinates, warp reference features according to the estimated motion, and decode the warped features into RGB values. Consequently, the reconstruction quality of C-STVSR heavily depends on the accuracy of motion estimation.

To examine how existing C-STVSR methods model motion, we compare the motion predicted by existing C-STVSR with pseudo motion estimated by RAFT [18] using the ground-truth intermediate frames. As shown in Fig. 1 (b), the motions predicted by MoTIF and BF-STVSR fail to faithfully preserve object structures, leading to significant errors, especially around object boundaries. These results reveal that existing methods struggle to model temporally coherent and consistent motion.

As shown in Fig. 2, this limitation can be attributed to two fundamental issues: (a) Motion multiplicity and (b) Temporal discrepancy. Motion multiplicity arises because existing methods vary only a single scalar temporal variable t while keeping the reference feature maps fixed. Although multiple plausible motion trajectories with different velocities or directions may exist between two frames, controlling motion solely with t constrains the model to learn a one-dimensional temporal representation. As a result, the model tends to converge to an averaged motion pattern that minimizes the training loss, thereby collapsing diverse trajectories into a single smoothed path. Temporal discrepancy arises from estimating motion at each time step independently, without explicit temporal dependency. Since previous motion information is not propagated when estimating motion at time t, temporal continuity cannot be enforced, leading to temporally inconsistent motion estimates. These issues propagate to the reconstruction stage, causing structural degradation and temporal inconsistency in the reconstructed frames.

To address these issues, we propose C^2-STVSR (Correlation-guided C-STVSR), which incorporates a bidirectional 4D correlation volume into C-STVSR. Unlike existing approaches that rely solely on a scalar temporal value to estimate motion at each time step, our method uses the correlation volume to provide time-varying motion cues for the current time step and recursively propagate previous motion information across time. This allows the model to capture more precise motion trajectories while enforcing temporal consistency between adjacent queried time steps. In addition, our framework learns task-oriented motion for C-STVSR without relying on supervision derived from pre-trained optical flow networks.

Extensive experiments demonstrate that our method achieves high-quality reconstruction across diverse spatial and temporal resolutions. C^2-STVSR consistently outperforms a broad range of fixed-scale and continuous STVSR baselines on various benchmarks. These results highlight the effectiveness of the proposed correlation-guided recursive motion modeling for continuous spatial-temporal video super-resolution.

2. Related Work

2.1. Space-Time Video Super-Resolution

Spatial-Temporal Video Super-Resolution (STVSR) jointly enhances spatial and temporal resolutions by reconstructing high-resolution intermediate frames from low-resolution inputs. Early STVSR methods proposed unified frameworks for joint spatio-temporal enhancement. Haris et al. [7] introduced an end-to-end STVSR architecture, and subsequent approaches incorporated bidirectional deformable ConvLSTM modules [20], refined temporal modeling strategies [21], and scale-aware feature aggregation [9] to better integrate spatial and temporal information. However, these methods are designed for fixed spatial and temporal scaling factors. To support arbitrary scaling in both dimensions, Continuous STVSR (C-STVSR) was introduced [4, 5, 11]. VideoINR [5] formulates STVSR as a continuous mapping from spatio-temporal coordinates to RGB values via implicit neural representations. MoTIF [4] improves motion-guided feature propagation through forward warping, while BF-STVSR [11] models continuous spatio-temporal representations using basis functions. Despite these advances, existing C-STVSR methods still have limited ability to model precise motion trajectories and temporal dependencies across arbitrary time steps.

3. Method

3.1. Overview

The overall architecture of the proposed method, C^2-STVSR, is illustrated in Fig. 3. Our framework is built upon the pipeline of MoTIF [4] and, inspired by LTE [12], applies Fourier representations to both temporal and spatial axes. Unlike prior approaches, we introduce bidirectional 4D correlation to enable more precise motion estimation, which allows the network to learn task-oriented motion for C-STVSR without relying on any external motion priors.

Given two low-resolution RGB video frames $I_0^L, I_1^L \in \mathbb{R}^{3\times H\times W}$ with spatial resolution $H \times W$, our objective is to synthesize a high-resolution intermediate frame $I_t^H \in \mathbb{R}^{3\times H'\times W'}$ at an arbitrary temporal coordinate $t \in [0,1]$, where the upscaling factor is defined as $s = \frac{W'}{W} = \frac{H'}{H} \geq 1$. To explicitly disentangle contextual information for super-resolution from motion information for motion estimation, we adopt two independent encoders: a context encoder E_C and a motion encoder E_M. The context encoder E_C takes the low-resolution frames as input and produces three latent features, $C_0^L, C_{(0,1)}^L, C_1^L \in \mathbb{R}^{C\times H\times W}$, which are used to generate the final high-resolution representation. Here, C_0^L and C_1^L denote the context features of I_0^L and I_1^L,109 respectively, while $C_{(0,1)}^L$ aggregates information from both input frames and serves as a template for the intermediate frame. Similarly, the motion encoder E_M extracts two latent features, $M_0^L, M_1^L \in \mathbb{R}^{C\times H\times W}$, representing the motion information of I_0^L and I_1^L, respectively. These motion features are used to construct bidirectional 4D correlation volumes that provide time-varying motion cues.

Based on the extracted features, the high-resolution context features and motion are predicted as follows: (1) Spatial-INR (S-INR): Given C_0^L and C_1^L as inputs, the S-INR generates high-resolution context features $C_0^H, C_1^H \in \mathbb{R}^{C\times sH\times sW}$ for a scaling factor s. (2) Correlation-Guided Temporal INR (CT-INR): Using M_0^L and M_1^L, a bidirectional 4D correlation volume is constructed. The CT-INR then predicts high-resolution motion $F_{0\to t}^H, F_{1\to t}^H \in \mathbb{R}^{3\times sH\times sW}$ at an arbitrary time step t and scale s using the correlation volume together with previously estimated motion information. Finally, the high-resolution context features are temporally propagated to the target time t via forward warping [16] based on the predicted motion $F_{0\to t}^H$ and $F_{1\to t}^H$, producing the target-time contextual feature C_t^H. The warped features are concatenated with target time t and $C_{(0,1)}^H$, a nearest-neighbor upsampled $C_{(0,1)}^L$, and decoded to reconstruct the final high-resolution interpolated frame I_t^H.

3.2. Correlation-Guided Temporal INR

Previous C-STVSR approaches adopt coordinate-based MLPs to model motion via implicit neural representations, enabling flexible prediction at arbitrary time steps t and scales s. However, predicting motion independently by varying only the scalar temporal variable t often fails to capture diverse and temporally coherent motion patterns. To address this limitation, we incorporate bidirectional 4D correlation between the two reference frames to learn time-varying motion representations and further introduce a recursive update scheme based on previous motion states to enforce temporally consistent motion modeling across time.

Bidirectional 4D correlation volumes. We capture dense correspondences between the reference frames by computing inner-product similarities between all pairs of motion features M_0^L and M_1^L to construct a 4D correlation volume. Since synthesizing an intermediate frame requires modeling motion from both reference frames toward the target time step, we construct the volume bidirectionally rather than unidirectionally. This 4D formulation encodes dense pairwise similarities between spatial locations of the two reference frames, providing rich correspondence cues for subsequent motion modeling.

Given two motion features M_0^L and M_1^L, the correlation volume for each direction $(a,b) \in \{(0,1),(1,0)\}$ is computed:

$$S_{ijkl}^{a\to b} = \sum_h M_{a,ijh}^L \cdot M_{b,klh}^L, \quad S^{a\to b} \in \mathbb{R}^{H\times W\times H\times W}. \tag{1}$$

To capture similarity information at multiple spatial scales, we progressively downsample the last two dimensions of each bidirectional correlation volume $S^{a\to b}$ using successive 2D average pooling operations with a kernel size of 2 and stride 2. This hierarchical downsampling constructs a three-level bidirectional correlation pyramid $\{S_\ell^{a\to b}\}_{\ell=0}^2$.

Correlation-guided time-varying motion cues. The bidirectional correlation volumes are queried to extract time-varying motion cues for predicting bidirectional motion at a target time step $t_k \in [0,1]$. To locate the query position at t_k, we propagate each pixel coordinate from

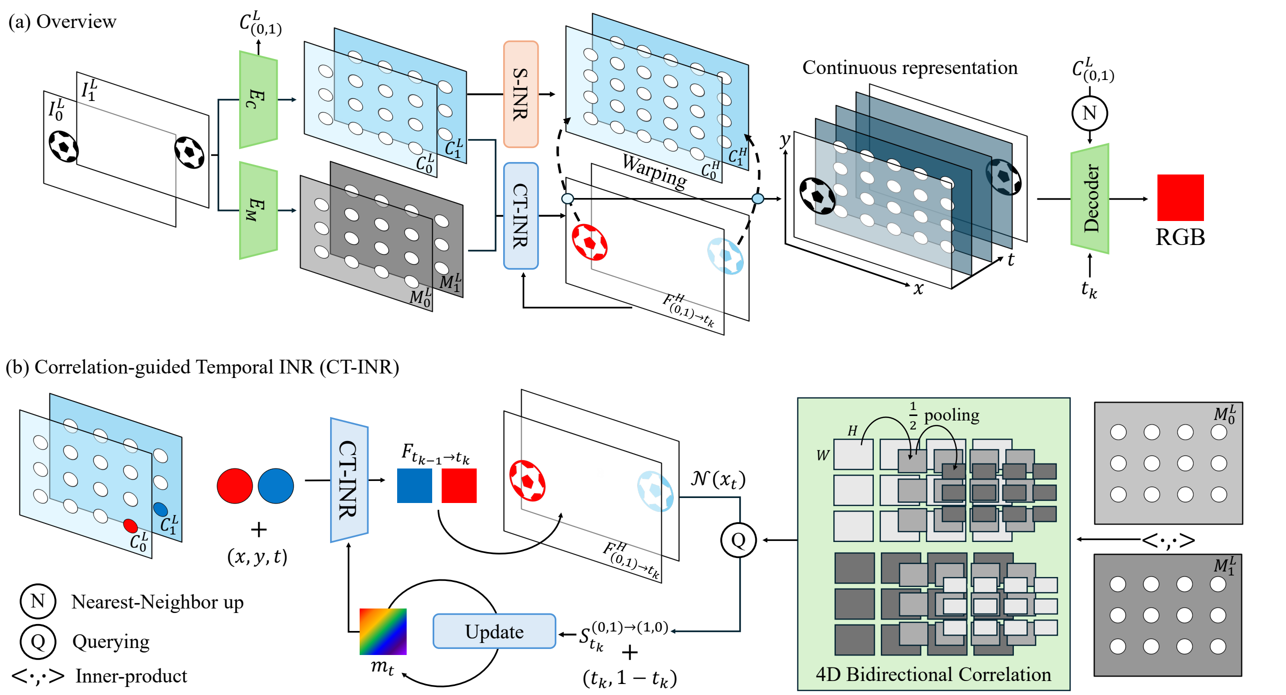


Figure 3. **Overview of the proposed C^2-STVSR**. (a) Input frames are encoded into low-resolution context and motion features, which are fed into S-INR and CT-INR to predict high-resolution contextual features and motion at each time t. The contextual features are temporally propagated by warping with the predicted motions. The resulting continuous representations are decoded to generate high-resolution interpolated RGB frames. (b) **CT-INR:** Motion features are used to construct bidirectional 4D correlation volumes. Given low-resolution contextual features, the motion representation, and spatial coordinates, CT-INR predicts high-resolution motion at time t. The predicted motion is accumulated over time and used to query the correlation volumes to update the motion representation. The updated motion representation serves as the input for predicting motion at the next time step.

the previous time step using the motion predicted at t_{k-1}. Specifically, each pixel coordinate $x_{t_{k-1}}$ is projected to the next time step as $x_{t_k} = x_{t_{k-1}} + F^L_{t_{k-2}\to t_{k-1}}(x_{t_{k-1}})$, where $F^L_{t_{k-2}\to t_{k-1}}$ denotes the downsampled motion estimated at time t_{k-1}. We define a local neighborhood around x_{t_k} within a predefined radius r as:

$$\mathcal{N}(x_{t_k}) = \left\{ x_{t_k} + \Delta x \;\middle|\; \Delta x \in \mathbb{Z}^2,\ \|\Delta x\|_1 \le r \right\}. \quad (2)$$

For notational simplicity, we describe the process only for the $(0, 1)$ direction; the same operation is applied to the $(1, 0)$ direction to ensure bidirectional motion modeling. Using the local neighborhood $\mathcal{N}(x_{t_k})$, we query the bidirectional correlation volume at the projected coordinates to extract motion-aware cues, denoted by $S^{a\to b}_{t_k}$. This operation is performed over all three levels of the correlation pyramid, where the neighborhood is scaled according to the spatial resolution of each level. The three-level correlation pyramid, together with the neighborhood radius $r = 2$, is designed according to the training patch size to provide sufficient multi-scale motion coverage within the given spatial.

Temporally consistent motion modeling. The correlation cue $S^{a\to b}_{t_k}$ is used to update the motion representation at time step t_k. At each time step t_k, the update block takes as input the temporal information $(t_k, 1-t_k)$, previous motion state $m_{t_{k-1}}$, the correlation cue $S^{a\to b}_{t_k}$, and the accumulated motion $F^L_{a\to t_{k-1}} = \sum_{i=1}^{k-1} \Delta F^L_{t_{i-1}\to t_i}$, and produces the updated motion latent m_{t_k}. The motion latent m_{t_k} is fed into the CT-INR to generate the high-resolution motion at time t_k. Here, $\delta_r = q - q_r$ denotes the spatial relative coordinate between the query coordinate $q = (x, y)$ and its nearest reference coordinate q_r on the low-resolution feature grid.

$$F^H_{t_{k-1}\to t_k} = \text{CT-INR}(m_{t_k}, \delta_r, t_k).$$

The predicted motion $F^H_{t_{k-1}\to t_k}$ is downsampled to the low-resolution as $F^L_{t_{k-1}\to t_k}$, which is then used to update the query neighborhood $\mathcal{N}(x_{t_{k+1}})$ and to recursively accumulate motion toward the subsequent time step t_{k+1}. Through this recursive process, motion estimation proceeds sequentially over time steps $\{t_0, t_1, \ldots\}$. Finally, the estimated motions are applied via softmax splatting to warp the high-resolution context features C^H_0 and C^H_1, producing the target-time contextual feature C_t^H for final frame reconstruction.

3.3. Training Objective

The proposed model is trained end-to-end without relying on any pre-trained optical flow distillation loss. While prior approaches [4, 11] employ an additional motion supervision term defined as $\mathcal{L}_{\text{RAFT}} = \sum_{i=0}^{1} \mathcal{L}_{\text{Char}}\left(\hat{F}_{i\to t}^H, F_{i\to t}^H\right)$, where $\mathcal{L}_{\text{Char}}$ denotes the Charbonnier loss, and $\hat{F}_{i\to t}^H$ and $F_{i\to t}^H$ denote the motion predicted by RAFT and the model, respectively. In contrast, we optimize the model using only the reconstruction objective:

$$\mathcal{L} = \mathcal{L}_{\text{Char}}\left(I_t^H, \hat{I}_t^H\right) = \sqrt{\left\| I_t^H - \hat{I}_t^H \right\|_2^2 + \epsilon^2}, \quad (3)$$

where I_t^H is the ground-truth high-resolution frame at time t, and $\hat{I}_t^H$ is the reconstructed frame.

4. Experiments

4.1. Experimental Setup

Implementation details. Unless otherwise stated, our implementation follows VideoINR [5]. Since previous C-STVSR methods were reported under different experimental settings, including training data preparation and batch size, we reproduce the C-STVSR baselines (VideoINR [5], MoTIF [4], and BF-STVSR [11]) under the same settings as VideoINR for fair comparison. We employ a two-phase training scheme where the spatial scaling factor is fixed at 4 during the initial 450k iterations and subsequently sampled from a uniform distribution over $[2, 4]$. Optimization is performed using Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with a cosine annealing schedule that reduces the learning rate from 1×10^{-4} to 1×10^{-7} every 150k iterations. We train the model with a batch size of 24, employing random rotations and horizontal flips for data augmentation. To enhance stability, we gradually anneal the probability of substituting predicted forward motion with ground-truth motion from 1.0 to 0 over the first 150k iterations. Architecture details are provided in the supplementary material.

Datasets. We train our model on the Adobe240 dataset [17], which contains 133 handheld 720p video sequences. All data preprocessing procedures follow those of VideoINR [5]. During training, we sample nine consecutive frames from each video clip, with the first and ninth frames serving as reference inputs. Three frames are randomly drawn and used as supervision targets. For evaluation, we adopt the Vid4 [14], Adobe240 [17], GoPro [15], and Vimeo-90K [23] datasets. Unless otherwise specified, the default spatial and temporal scaling factors are set to 4 and 8, respectively. For Adobe-*Center*, GoPro-*Center*, Vimeo-`Fast`, Vimeo-`Medium`, and Vimeo-`Slow`, we conduct evaluation using only the 1st, 4th, and 9th frames, which correspond to the center-frame interpolation setting.

Baselines and evaluation metrics. We categorize baseline methods into two types—fixed and continuous-scale—and conduct comparisons within each category. Fixed-scale Spatial-Temporal Video Super-Resolution (Fixed-STVSR) methods operate at predefined spatial and temporal scaling factors determined during training. For two-stage Fixed-STVSR, we combine fixed video super-resolution models (bicubic interpolation, EDVR [19], and BasicVSR [2]) with video frame interpolation models (SuperSloMo [10], QVI [22], and DAIN [1]). As a representative one-stage Fixed-STVSR method, we include ZoomingSlowMo [20]. For C-STVSR, we consider two-stage pipelines that combine continuous image super-resolution models (LIIF [3] and LTE [12]) with frame interpolation models (RIFE [8] and EMA-VFI [24]). We also compare against one-stage continuous methods, including TMNet [21] (limited to $\times 4$ spatial scaling), VideoINR [5], MoTIF [4], and BF-STVSR [11]. We report Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) on the Y channel. We further adopt FloLPIPS [6], a perceptual metric originally designed for video frame interpolation, to measure temporal consistency, and VMAF [13], a perceptual quality metric widely used in video streaming, to evaluate overall video fidelity.

4.2. Quantitative Results

We compare our proposed method with Fixed-STVSR methods in Tab. 1. For single-frame interpolation settings, including Vid4, GoPro-*Center*, and Adobe-*Center*, our method achieves the best performance on all datasets except Vid4. On Vid4, TMNet demonstrates relatively strong performance compared with other methods. This may be attributed to its training on the Vimeo-90K dataset [23], which shares similar characteristics with Vid4, as well as its subsequent fine-tuning on Adobe240 data. For multi-frame interpolation settings, represented by GoPro-*Average* and Adobe-*Average*, our method also outperforms prior approaches. We believe these gains mainly stem from our correlation-guided recursive motion modeling, which conditions current motion estimation on previous motion states and leverages bidirectional correlation volumes to learn time-varying motion representations. As a result, our method achieves more precise and temporally coherent motion estimation, leading to more stable and accurate frame reconstruction.

Tab. 2 compares the performance of C-STVSR methods on the GoPro dataset under out-of-distribution scaling factors. The compared methods include both two-stage and one-stage C-STVSR approaches. C^2-STVSR achieves the

Table 1. Comparison with Fixed-STVSR baselines on the Vid4, GoPro, and Adobe240 datasets. Performance is reported in terms of PSNR (dB) and SSIM. All results are obtained under $\times 4$ spatial and $\times 8$ temporal scaling. "*Average*" indicates the mean performance over all 8 interpolated frames, whereas "Center" corresponds to evaluation on the 1st, 4th, and 9th frames (i.e., center-frame interpolation). † indicates results reproduced under the same experimental settings as VideoINR [5]. The best results are marked in red.

VFI Method	VSR Method	Vid4	GoPro-*Center*	GoPro-*Average*	Adobe-*Center*	Adobe-*Average*	Parameters (Millions)
SuperSloMo [10]	Bicubic	22.42 / 0.5645	27.04 / 0.7937	26.06 / 0.7720	26.09 / 0.7435	25.29 / 0.7279	19.8
SuperSloMo [10]	EDVR [19]	23.01 / 0.6136	28.24 / 0.8322	26.30 / 0.7960	27.25 / 0.7972	25.90 / 0.7682	19.8+20.7
SuperSloMo [10]	BasicVSR [2]	23.17 / 0.6159	28.23 / 0.8308	26.36 / 0.7977	27.28 / 0.7961	25.94 / 0.7679	19.8+6.3
QVI [22]	Bicubic	22.11 / 0.5498	26.50 / 0.7791	25.41 / 0.7554	25.57 / 0.7324	24.72 / 0.7114	29.2
QVI [22]	EDVR [19]	23.48 / 0.6547	28.60 / 0.8417	26.64 / 0.7977	27.45 / 0.8087	25.64 / 0.7590	29.2+20.7
QVI [22]	BasicVSR [2]	23.15 / 0.6428	28.55 / 0.8400	26.27 / 0.7955	26.43 / 0.7682	25.20 / 0.7421	29.2+6.3
DAIN [1]	Bicubic	22.57 / 0.5732	26.92 / 0.7911	26.11 / 0.7740	26.01 / 0.7461	25.40 / 0.7321	24.0
DAIN [1]	EDVR [19]	23.48 / 0.6547	28.58 / 0.8417	26.64 / 0.7977	27.45 / 0.8087	25.64 / 0.7590	24.0+20.7
DAIN [1]	BasicVSR [2]	23.43 / 0.6514	28.46 / 0.7966	26.43 / 0.7966	26.23 / 0.7725	25.23 / 0.7725	24.0+6.3
ZoomingSloMo [20]		25.72 / 0.7717	30.69 / 0.8847	- / -	30.26 / 0.8821	- / -	11.10
TMNet [21]		25.96 / 0.7803	30.14 / 0.8696	28.83 / 0.8514	29.41 / 0.8524	28.30 / 0.8354	12.26
VideoINR† [5]		25.48 / 0.7645	30.09 / 0.8762	29.48 / 0.8688	29.60 / 0.8678	29.12 / 0.8621	11.31
MoTIF† [4]		25.51 / 0.7670	30.45 / 0.8822	29.58 / 0.8722	29.89 / 0.8761	29.18 / 0.8670	12.55
BF-STVSR† [11]		25.62 / 0.7696	30.65 / 0.8845	29.81 / 0.8751	30.15 / 0.8793	29.55 / 0.8722	13.47
C^2-STVSR (Ours)		25.72 / 0.7710	30.72 / 0.8847	29.90 / 0.8758	30.35 / 0.8816	29.70 / 0.8746	16.83

Table 2. Comparison with C-STVSR baselines for out-of-distribution scales on GoPro dataset. Performance is reported in terms of PSNR (dB) and SSIM. All results are obtained by a scaling factor specified in the table and metrics are calculated across all interpolated frames. † indicates results reproduced under the same experimental settings as VideoINR [5]. The best results are marked in red.

Temporal Scale	Spatial Scale	RIFE [8] LIIF [3]	RIFE [8] LTE [12]	EMA-VFI [24] LIIF [3]	EMA-VFI [24] LTE [12]	VideoINR† [5]	MoTIF† [4]	BF-STVSR† [11]	C^2-STVSR (Ours)
$\times 8$	$\times 4$	29.14 / 0.8524	29.14 / 0.8524	29.68 / 0.8671	29.68 / 0.8667	29.48 / 0.8688	29.58 / 0.8722	29.81 / 0.8751	29.90 / 0.8757
$\times 6$	$\times 4$	30.16 / 0.8738	30.16 / 0.8737	30.64 / 0.8850	30.64 / 0.8848	30.83 / 0.8968	31.03 / 0.9015	31.21 / 0.9034	31.38 / 0.9218
	$\times 6$	27.50 / 0.7900	27.45 / 0.7880	27.80 / 0.8000	27.75 / 0.7980	28.86 / 0.8397	28.97 / 0.8455	29.08 / 0.8465	29.22 / 0.8480
	$\times 8$	26.30 / 0.7500	26.25 / 0.7480	26.60 / 0.7600	26.55 / 0.7580	26.65 / 0.7841	26.58 / 0.7884	26.56 / 0.7889	26.71 / 0.7891
$\times 10$	$\times 4$	28.20 / 0.8305	28.20 / 0.8304	28.73 / 0.8468	28.73 / 0.8466	28.38 / 0.8428	28.38 / 0.8442	28.70 / 0.8491	28.69 / 0.8481
	$\times 6$	26.69 / 0.7766	26.69 / 0.7761	27.04 / 0.7884	27.05 / 0.7878	27.29 / 0.8027	27.23 / 0.8057	27.49 / 0.8090	27.52 / 0.8093
	$\times 8$	18.80 / 0.6039	18.85 / 0.6051	18.77 / 0.6042	18.83 / 0.6059	25.74 / 0.7596	25.58 / 0.7617	25.70 / 0.7649	25.76 / 0.7635

Table 3. Comparison of one-stage C-STVSR methods on the Adobe240 dataset using FloLPIPS [6] and VMAF [13]. The best results are shown in red.

Method	VideoINR	MoTIF	BF-STVSR	Ours
FloLPIPS↓	0.207	0.204	0.199	0.198
VMAF↑	51.48	53.16	54.16	54.56

best performance in most scaling scenarios, suggesting that the proposed correlation-guided time-varying motion representation enables robust motion modeling across varying spatial and temporal scales. We also compare with one-stage C-STVSR methods using perceptual video quality metrics in Tab. 3. Our method achieves the best VMAF score and a comparable FloLPIPS score to BF-STVSR, demonstrating strong perceptual quality and temporal consistency.

To further analyze robustness under different motion magnitudes, we evaluate C-STVSR methods on motion-

Table 4. Comparison with C-STVSR baselines on the Vimeo-90K. Results are reported in PSNR (dB) and SSIM. The best results are shown in red.

Method	Vimeo-`Fast`	Vimeo-`Medium`	Vimeo-`Slow`
VideoINR	30.77 / 0.8649	31.16 / 0.8840	30.60 / 0.8824
MoTIF	31.26 / 0.8598	31.36 / 0.8860	30.53 / 0.8818
BF-STVSR	31.64 / 0.8694	31.59 / 0.8880	30.73 / 0.8835
Ours	31.67 / 0.8695	31.76 / 0.8897	30.90 / 0.8856

partitioned subsets of the Vimeo-90K test set. Following [20], the test sequences are divided into three groups: `Fast`, `Medium`, and `Slow`. As shown in Tab. 4, our method consistently achieves higher PSNR across all motion categories than prior approaches.

4.3. Qualitative Results

Fig. 4 shows the estimated motion and interpolated results at $t = 0.5$ under $\times 4$ spatial and $\times 6$ temporal interpolation,

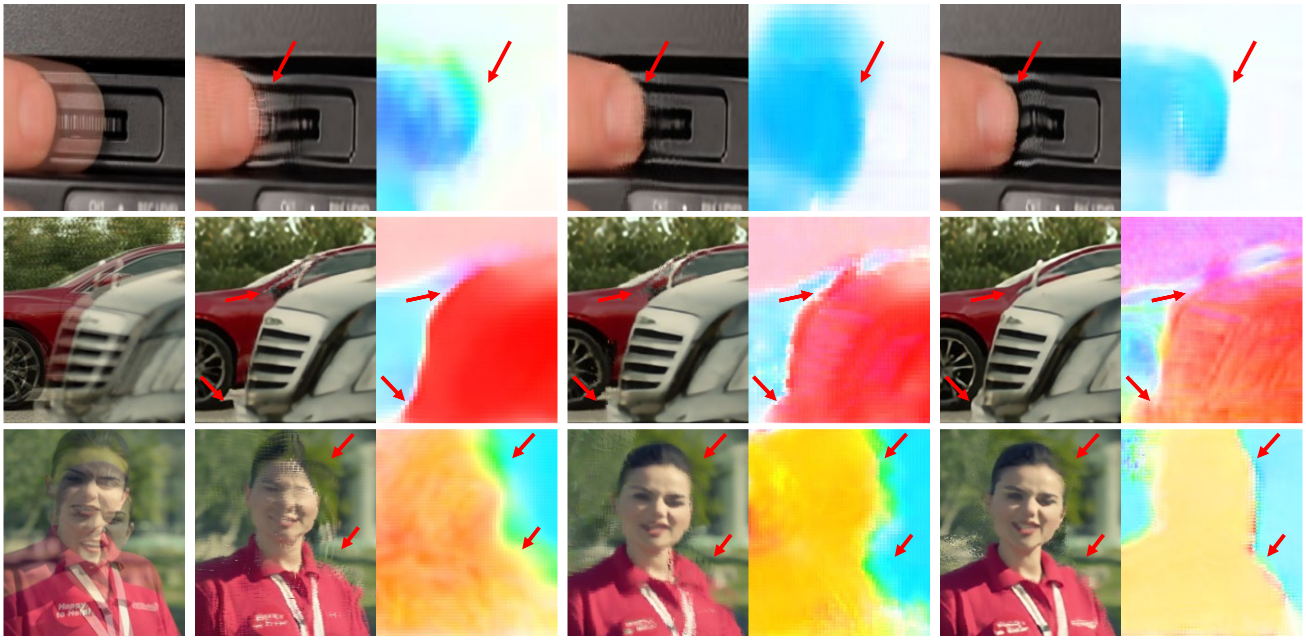


Figure 4. Qualitative comparison for $\times 4$ spatial and $\times 6$ temporal interpolation. "Overlap" denotes the averaged image of the two input frames ($t = 0, 1$), and the following images show the interpolated results and estimated motions $F^{H}_{0\to t}$ at $t = 0.5$.

where the visualized motion represents the forward motion from the reference frame at $t = 0$ to the intermediate frame. Existing methods often estimate inaccurate motion, leading to noticeable blur or distorted structures. In contrast, C^2-STVSR produces more reliable motion estimates and clearer interpolated frames. Fig. 5 presents qualitative comparisons with VideoINR, MoTIF, and BF-STVSR at $t = 0.5$ under $\times 4$ spatial and $\times 8$ temporal interpolation. The examples are grouped into Large, Medium, and Small cases according to object scale to analyze reconstruction performance across different sizes. In the Large case, other methods often produce distorted structures around moving subjects, whereas C^2-STVSR reconstructs the students more clearly while better preserving their shapes. In the Medium case, other methods show blur or ghosting artifacts around the person and bicycle, whereas C^2-STVSR preserves their shapes and motion details with sharper reconstructions. In the Small case, other methods produce over-smoothed results, whereas C^2-STVSR restores details such as the hand and wristwatch while maintaining structural consistency. Overall, these results show that C^2-STVSR effectively handles motion across different object scales while preserving structural integrity and fine details.

4.4. Ablation Study

Ablation study on key components. As shown in Tab. 5, we analyze the contributions of two key components: the bidirectional 4D correlation volumess (Corr) and temporally consistent motion modeling (Recur). Each component individually improves performance on both the GoPro and Adobe240 datasets, indicating that Corr provides useful spatial correspondence cues for motion estimation, while Recur enhances temporal consistency across time steps. When both components are combined, the model achieves the best PSNR and SSIM across all evaluation settings, improving PSNR by up to 0.34 dB over the baseline, confirming that the two components are complementary and jointly contribute to accurate motion modeling in C-STVSR.

Table 5. Ablation study on the key components of C^2-STVSR: the bidirectional 4D correlation volumes (Corr) and temporally consistent motion modeling (Recur). Results are reported in PSNR (dB) and SSIM. The best results are marked in red.

Corr	Recur	GoPro-*Average*	GoPro-*Center*	Adobe-*Average*	Adobe-*Center*
		29.69 / 0.8739	30.47 / 0.8830	29.42 / 0.8711	30.01 / 0.8780
✓		29.73 / 0.8738	30.45 / 0.8823	29.51 / 0.8713	30.08 / 0.8780
	✓	29.79 / 0.8751	30.61 / 0.8843	29.55 / 0.8726	30.17 / 0.8799
✓	✓	29.90 / 0.8757	30.72 / 0.8847	29.70 / 0.8746	30.35 / 0.8816

Task-oriented motion learning. Tab. 6 presents the ablation results on knowledge distillation (KD) using a pretrained optical flow network. We compare MoTIF [4], BF-STVSR [11], and the proposed C^2-STVSR with and without $\mathcal{L}_{RAFT}$. MoTIF, which employs RAFT for motion feature extraction and supervision, shows the lowest performance among the compared methods. For BF-STVSR, introducing $\mathcal{L}_{RAFT}$ leads to marginal or even negative performance changes. This observation suggests a mismatch

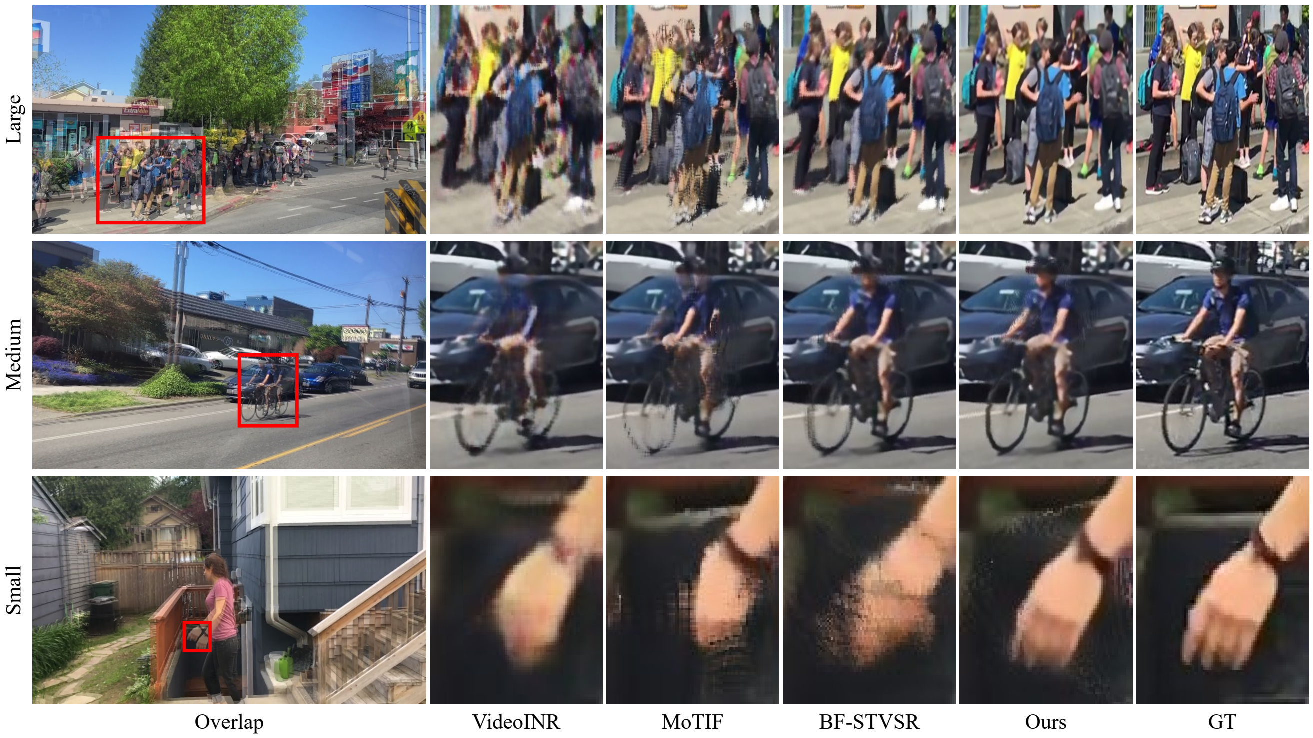


Figure 5. Qualitative comparison for $\times 4$ spatial and $\times 8$ temporal interpolation. "Overlap" denotes the averaged image of the two input frames ($t = 0, 1$), and the following images show the interpolated results at $t = 0.5$. The scenes are categorized into Large, Medium, and Small based on the object size within the frames.

Table 6. The impact of knowledge distillation from a pre-trained optical flow network. $\mathcal{L}_{RAFT}$ denotes supervision using RAFT [18]. Results are reported in PSNR (dB) and SSIM. † indicates results reproduced under the same experimental settings as VideoINR [5]. The best results are marked in red.

Method	$\mathcal{L}_{RAFT}$	GoPro-*Center*	GoPro-*Average*	Adobe-*Center*	Adobe-*Average*
MoTIF†	✓	30.45 / 0.8822	29.58 / 0.8722	29.89 / 0.8761	29.18 / 0.8670
BF-STVSR†	✓	30.62 / 0.8842	29.78 / 0.8748	30.20 / 0.8806	29.57 / 0.8735
BF-STVSR†		30.65 / 0.8845	29.81 / 0.8751	30.15 / 0.8793	29.55 / 0.8722
C^2-STVSR	✓	30.34 / 0.8801	29.68 / 0.8724	29.83 / 0.8736	29.29 / 0.8678
C^2-STVSR		30.72 / 0.8847	29.90 / 0.8758	30.35 / 0.8816	29.70 / 0.8746

between the objectives of optical flow and C-STVSR: optical flow estimates discrete inter-frame displacement between observed frames, whereas C-STVSR requires modeling motion continuously at unobserved intermediate time steps. In contrast, C^2-STVSR adopts correlation-guided recursive motion modeling and is trained end-to-end for the target task. As a result, the learned motion naturally aligns with the objective of C-STVSR, achieving the best performance without external flow supervision. Notably, the performance degradation observed when $\mathcal{L}_{\text{RAFT}}$ is introduced suggests that supervision not aligned with the C-STVSR objective may interfere with learning motion representations specialized for the task. These findings indicate that C^2-STVSR can effectively learn task-oriented motion representations in an end-to-end manner.

5. Conclusion

In this paper, we propose C^2-STVSR, a novel framework for continuous spatial-temporal video super-resolution (C-STVSR). We identify two underexplored challenges in C-STVSR, motion multiplicity and temporal discrepancy, and address them through correlation-guided recursive motion modeling with bidirectional 4D correlation volumes. By constructing a time-varying motion representation conditioned on previous motion information, the proposed framework enables accurate and temporally consistent motion estimation. Moreover, C^2-STVSR learns task-oriented motion for C-STVSR without relying on supervision from pre-trained optical flow networks. Extensive experiments confirm that C^2-STVSR consistently outperforms existing approaches in PSNR, SSIM, and perceptual video quality metrics, demonstrating clear spatial detail, coherent temporal reconstruction, and robustness under challenging conditions, including out-of-distribution scaling factors.

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